

Query Optimization: Transformation of Relational Expressions DBMS

> **Definition:** Query Optimization is the component of a Database Management System (DBMS) that attempts to determine the most efficient execution plan for evaluating a given query by transforming relational algebra expressions into equivalent, lower-cost evaluation trees.

1. Detailed Technical Explanation

Query Processing Steps:

High-Level Overview:

The query processing steps involve several key phases:

- 1. **Query Parsing:** The query is parsed into a relational algebra expression.
- 2. **Query Rewrite Rules:** These rules are used to transform the query into an equivalent form that is easier to optimize.
- 3. **Query Optimization:** The query is transformed into a sequence of relational algebra operations that can be executed efficiently.
- 4. **Query Execution:** The query is executed using the optimized plan.

Key Relational Algebra Equivalence Rules:

1. **Commutativity of Selection:**
 $\sigma_{A \wedge B}(R) = \sigma_{B \wedge A}(R)$
2. **Associativity of Selection:**
 $\sigma_{A \wedge B \wedge C}(R) = \sigma_{A \wedge (\sigma_{B \wedge C}(R))}$
3. **Distributivity of Selection over Join:**
 $\sigma_{A \wedge B}(R \bowtie S) = (\sigma_{A \wedge B}(R) \bowtie S)$
4. **Projection Elimination:**
 $\pi_{A, B}(\pi_{A, B, C}(R)) = \pi_{A, B}(R)$
5. **Join Elimination:**
 $\pi_{A, B}(\sigma_{A \wedge B}(R \bowtie S)) = (\pi_{A, B}(R) \bowtie S)$

Heuristic Optimization Algorithm:

1. Break down complex query conditions into simple selections.
 2. **Projection Pushdown:** Push projections down to the data sources to reduce the amount of data processed.
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2. Core Concepts & Memory Keywords

- **Equivalence Rules:** Algebraic identities ensuring two query trees return identical tuple results.
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3. Must-Write Points for Exams

- Query optimization chooses the physical execution path with minimal disk I/O cost.

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4. Quick Recall Flow

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